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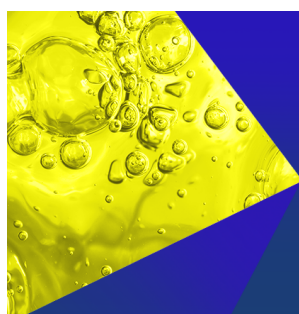
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Raj Gandhi,^{1,a)} Alexander Nepomnyashchy,¹ and Alexander Oron²

AFFILIATIONS

¹Department of Mathematics, Technion—Israel Institute of Technology, Haifa 3200003, Israel

²Department of Mechanical Engineering, Technion—Israel Institute of Technology, Haifa 3200003, Israel

^{a)}Author to whom correspondence should be addressed: raj.gandhi@campus.technion.ac.il

ABSTRACT

We investigate the thermosolutal Marangoni instability in a moderately dense nanoparticle suspension (nanofluid). The nanofluid layer deposited on a solid substrate and subjected to a specified constant heat flux is exposed to the ambient atmosphere at the air–liquid deformable interface. The effects of thermophoresis and gravity are considered. Using the method of asymptotic expansions, we investigate the oscillatory long-wave thermosolutocapillary instabilities, taking into consideration the dependence of the thermophysical properties of the nanofluid on the local nanoparticle concentration. Variations in the threshold of the long-wave oscillatory thermocapillary instability with the heat transfer rate at the interface and the interface deformability are obtained. We find a transition from monotonic to oscillatory thermosolutal Marangoni instability with a codimension-two point bifurcation. An excellent agreement is obtained between the results of the asymptotic analysis and those of the numerical solution of the full eigenvalue problem, in terms of the critical values of the Marangoni number, the wavenumber, and the frequency.

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I. INTRODUCTION

Surface tension-driven phenomena play a major role around us.^{1–5} Inhomogeneities of the scalar fields such as temperature, solute or particle concentration, and electric charge, at the layer interface result in nonuniformity of surface tension along the interface, thereby creating a driving force to an instability known as *Marangoni* instability,⁶ and the respective variations may lead to thermocapillary, solutocapillary, and electrocapillary instability.⁷ In a non-deformable liquid layer, Pearson⁸ found the monotonic Marangoni instability induced by either the uniform temperature or by application of the heat flux at the solid substrate underneath. Later, Scriven and Sternling⁹ found that the interfacial deformation plays an important role in the stability properties of a system. Further, Smith¹⁰ corrected the zero value for the critical Marangoni number⁹ and showed the stabilization of the long-wavelength gravitational waves.

It should be emphasized that, in certain situations, the Marangoni effect is purposefully employed, for instance, for mixing in the low-gravity environment, capillary pumping, bubble removal in glass melts, suppression of the Rayleigh–Taylor instability, and prevention of radial segregation in semiconductor crystal growth.^{7,11–14}

Thermophoresis, or the Soret effect, is known to induce the concentration gradient by imposing a temperature gradient. In binary mixtures, the Soret effect introduces an additional component in the mass flux associated with the temperature field. This coupling between the concentration and temperature fields significantly affects the stability properties of the system.^{15,16} Oron and Nepomnyashchy¹⁷ and Joo¹⁸ found the onset of both monotonic and oscillatory instabilities in a layer of a binary mixture with either a non-deformable or deformable interface. Joo¹⁸ reported the emergence of oscillatory instability when the solutocapillarity stabilizes the system, whereas the thermocapillarity destabilizes it. The resulting thermosolutocapillary instability emerges as a long-wavelength or a finite-wavelength instability depending on the system parameters.^{17,19–23} For a sufficiently low value of the Biot number B , which characterizes the amount of heat transfer to the atmosphere, it was found that for small values of B the instability tends to be long-wave. Podolny *et al.*¹⁹ and Oron and Nepomnyashchy¹⁷ revealed two distinct asymptotic regimes of instability at the limits of the perturbation wavenumber $k \sim B^{1/4}$ and $k \sim B^{1/2}$.

In many technological applications, such as nanofluid fuels, ink-jet printing, material processing, and coating,^{24,25} nanometer-size

particles or colloids are suspended in carrier liquids. In a nanofluid,^{26,27} all its thermophysical properties, such as thermal conductivity, density, viscosity, and heat capacity, depend on the local concentration of the nanoparticles. Recently, we have investigated²⁸ the thermosolutocapillary instabilities emerging in a nanofluid layer subjected to heating or cooling at the supporting substrate. We found that the critical wavenumber k_c for the onset of instability may vary from low to finite values depending on the system parameters and may also display various complex competitions between these domains. In Gandhi *et al.*,²⁹ we studied the onset of long-wavelength monotonic instability in a nanofluid layer in the limit of a weakly deformable interface. We showed that the surface tension significantly affects the stability of a system and may modify the type of Marangoni instability. Moreover, we found that both monotonic and oscillatory thermosolutal instability may emerge. The investigation of the long-wave regime presented in Gandhi *et al.*²⁹ with $k_c \sim B^{1/4}$ and the dimensionless surface tension number $\Sigma_0 \sim O(1)$ related to the degree of deformability of the liquid–gas interface, successfully predicts threshold values of the Marangoni number associated with the monotonic mode of instability. However, it does not account for the effect of interface deformability.

The present paper investigates the effect of surface tension or the interface deformability on the instability threshold and the corresponding variation of the critical wavenumber in the low wavenumber regime. We employ the method of asymptotic expansions and fully numerical computations related to the corresponding eigenvalue problem to determine the instability threshold. The plan of the paper is as follows. Section II provides the details of the problem formulation, followed by the identification of the quiescent steady base state of the system. Section III introduces the linear stability analysis around the determined base state and presents the corresponding eigenvalue problem. Section IV is dedicated to the details of the long-wavelength expansion and the solution procedure. Section V presents the results for the nanofluid layer heated from below, with variation in surface tension, Biot number, and solutocapillarity. Finally, Sec. VI summarizes the findings of the paper.

II. PROBLEM FORMULATION AND GOVERNING EQUATIONS

We consider a two-dimensional (2D) nanofluid layer of a mean thickness h_0^* whose thermophysical properties are density ρ^* , dynamic viscosity μ^* , kinematic viscosity $\nu^* = \mu^*/\rho^*$, thermal conductivity K^* , heat capacity c^* , and thermal diffusivity $\kappa^* = K^*/\rho^*c^*$. In what follows, we denote the properties of the nanofluid, the base fluid, and the nanoparticles with subscripts *nf*, *bf*, and *np*, respectively. The nanofluid layer resting on a poorly conducting solid planar substrate is assumed to have a deformable interface, thus of variable thickness h^* , where it is exposed to the quiescent gas environment held at constant pressure p_∞^* and temperature T_∞^* in the gravity field g^* . We use the reference frame in which the x^* axis is located at the substrate, whereas the z^* axis is normal to the latter and directed into the fluid layer opposite to the direction of gravity. Thus, the substrate and the deformable liquid–gas interface are located at $z^* = 0$ and $z^* = h^*$, respectively, as illustrated in Fig. 1.

The nanofluid layer is assumed to be moderately dense with an average bulk volumetric nanoparticle concentration $\Phi^* \leq 0.3$, wherein the nanoparticle diameter $d_p^* \approx 10$ nm, and the carrier liquid is Newtonian. The local instantaneous volumetric nanoparticle

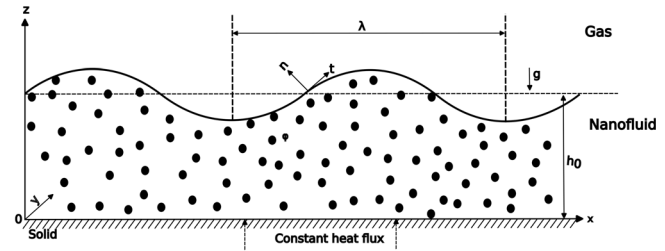


FIG. 1. Nanofluid ($d_p^* = 10 - 100$ nm) layer on a poorly conducting solid substrate subjected to a constant heat flux from the substrate and exposed to the gas phase at its deformable interface.

concentration is denoted by $\phi^*(x^*, z^*, t^*)$. The average solute concentration Φ^* is thus determined via

$$\Phi^* = \frac{1}{\mathcal{S}_l^* h_0^*} \int_0^{h^*} \left(\int_{\mathcal{S}^*} \phi^*(x^*, z^*, t^*) dx^* \right) dz^*, \quad (1)$$

where $\mathcal{S}^*$ is a projection of the flow domain onto the x^* axis and $\mathcal{S}_l^*$ is the length of this projection. We note that Φ^* is assumed to be constant in time due to the impermeability boundary conditions imposed on the system, see below. We note in passing that in the setting considered below, the base state is a quiescent steady state; therefore, Φ^* given by Eq. (1) is conserved. Would the system have base-state flow, the conservation of mass flux or lateral nanoparticle concentration flux

$$\Phi^* = \frac{1}{\mathcal{S}_l^* h_0^*} \int_0^{h^*} \left(\int_{\mathcal{S}^*} u^*(x^*, z^*, t^*) \phi^*(x^*, z^*, t^*) dx^* \right) dz^*$$

should have to be considered, which effectively takes into consideration the variation of the nanoparticle concentration across the layer.^{30–33}

The considered system is heated by the prescribed heat flux $q^* > 0$ at its solid substrate in the direction normal to the latter. The specified heat flux leads to the emergence of the temperature field within the layer that varies with time and space, $T^* = T^*(x^*, z^*, t^*)$. We also assume that the Soret effect expressing the mass flux as a combination of the temperature and particle concentration gradients is in effect.

For most pure liquids and binary mixtures, surface tension at the liquid–gas interface decreases with the interfacial temperature and concentration, except for inorganic salt, whose surface tension when dissolved in water increases with its interfacial concentration;¹⁷ therefore, for moderately dense nanosuspensions, it is adequately approximated by a linear function³⁴

$$\sigma^*(T^*, \phi^*) = \sigma_r^* - \sigma_{T^*}^*(T^* - T_r^*) - \sigma_{\phi^*}^*(\phi^* - \phi_r^*), \quad (2a)$$

with

$$\sigma_{T^*}^* = -\partial\sigma^*/\partial T^* > 0 \quad \text{and} \quad \sigma_{\phi^*}^* = -\partial\sigma^*/\partial\phi^* > 0, \quad (2b)$$

where σ_r^* is the equilibrium surface tension at the reference temperature T_r^* and interfacial particle concentration ϕ_r^* , so the surface tension linearly decreases with the temperature and particle concentration at the interface. We note that the dispersion of alumina nanoparticles in distilled water exhibits a negligible value of the adsorption/desorption coefficient ratio, and in this case, the interfacial kinetics mechanism may be neglected.³⁴ However, nanoparticle dispersions in various liquids, such as non-stabilized water, n-decane, n-dodecane, n-

hexadecane, and others, are significantly affected by interfacial kinetics. In these conditions, consideration of interfacial kinetics becomes necessary,³⁴ and the combined action of the bulk particle concentration ϕ^* and an interfacial particle concentration Γ^* must be taken into consideration in Eqs. (2a) and (2b).

The thermophysical properties of a dilute binary mixture, such as density, viscosity, thermal conductivity, and heat capacity, are fairly well approximated as constants. However, in the case of moderately dilute mixtures, their thermophysical properties strongly depend on the particle concentration.²⁶

The density and the specific heat capacity of the nanofluid are determined as the weighted sum of the respective properties of the base fluid and the nanoparticle volume fraction

$$\begin{aligned}\rho_{nf}^*(\phi^*) &= (1 - \phi^*)\rho_{bf}^* + \phi^*\rho_{np}^*, \\ c_{nf}^*(\phi^*) &= \frac{(1 - \phi^*)(\rho^*c^*)_{bf} + \phi^*(\rho^*c^*)_{np}}{\rho_{nf}^*}.\end{aligned}\quad (3a)$$

In the case of the nanofluid dynamic viscosity, we make use of the empirical correlation^{35–37} for a concentrated suspension given by

$$\mu_{nf}^*(\phi^*) = \mu_{bf}^* \left(1 - \frac{\phi^*}{\phi_m}\right)^{-2}, \quad (3b)$$

where ϕ_m is the maximal packing fraction taken in our computations as $\phi_m = 0.65$ which is related to the random close packing (RCP) volume fraction of nanoparticles. This model illustrates the power-law variation of the nanofluid viscosity with the nanoparticle concentration, diverging when the nanoparticle concentration becomes too high, $\phi^* \rightarrow \phi_m$.

The thermal conductivity of nanofluids is also found to strongly depend on the particle concentration, and for instance, for alumina Al_2O_3 nanoparticles in water, the empirical model is given by Buongiorno²⁶ based on the experimental data²⁷

$$K_{nf}^*(\phi^*) = K_{bf}^*(1 + a\phi^*), \quad (4)$$

where a is a constant describing the inhomogeneity of the thermal conductivity, $a = 7.47$. We note that empirical data for nanofluids are scarce, and in such cases, analytical expressions derived by Maxwell³⁸ and Jeffrey³⁹ can also be used in the limit of low nanoparticle concentration.

Among the seven different physical mechanisms relevant to nanoparticle motion in a nanofluid,²⁶ we consider two dominant ones, namely Brownian diffusion and the Soret effect, known also as thermophoresis, in a differentially heated fluid. For low-concentration suspensions, the Brownian diffusion coefficient is given by the generalized Stokes–Einstein formula^{40–42}

$$\mathcal{D}(\phi^*) = D_B(1 + 1.45\phi^*), \quad (5)$$

where $D_B = \frac{K_B T^*}{3\pi\mu_{bf}^* d_p^*}$ and $K_B = 1.380\,649 \times 10^{-23}$ J/K are, respectively, the diffusivity constant given by the Stokes–Einstein formula and the Boltzmann constant. However, we note that the Brownian diffusion coefficient for a nanoparticle suspension depends on the local concentration Eq. (5), in what follows, we will employ as a simplification a constant $\mathcal{D}(\phi^*) = D_B$ (see Sec. 2.3 in Gandhi *et al.*²⁸ for more details).

For a nonisothermal fluid layer, the temperature gradient across the layer induces another slip mechanism in a nanofluid, namely the Soret effect. The Soret or thermal diffusion coefficient

in a nanofluid is proportional to the particle concentration ϕ^* and is expressed by^{43–46}

$$D_T = \left(\frac{0.26K_{bf}^*}{2K_{bf}^* + K_{np}^*} \right) \frac{\mu_{bf}^*}{\rho_{bf}^*} \phi^*. \quad (6)$$

Therefore, we note that the kinematic viscosity, thermal diffusivity, and the Soret coefficient of a nanoparticle suspension depend on the local nanoparticle concentration.⁴⁷ The total nanoparticle mass flux is expressed, therefore, by the summation of separate mass fluxes due to the Brownian motion and Soret effect

$$\mathbf{j}_p = -\rho_p^* \left(C_B T^* \nabla^* \phi^* + C_T \phi^* \frac{\nabla^* T^*}{T^*} \right), \quad (7)$$

where $C_B = D_B/T^*$, $C_T = D_T/\phi^*$, and $\nabla^* = (\partial_{x^*}, \partial_{z^*})$ with subscripts x^* and z^* denoting partial differentiation with respect to the corresponding variable.

The heat flux in the nanofluid is approximately given by Fourier's law of heat conduction

$$\mathbf{j}_T = -K_{nf}^* \nabla^* T^*. \quad (8)$$

The Dufour effect is exceedingly weak in liquid mixtures^{15,17,23} and can be safely neglected in the case at hand. Other components of the heat flux may also be omitted, and we refer the reader to the relevant discussion in Gandhi *et al.*²⁸ We stress that for a thin nanofluid layer and for small nanoparticles ($d_p^* \approx 10$ nm), the gravitational settling is negligible. However, as noted by Chang and Ruo,⁴⁸ for thick nanofluid layers and for large nanoparticles, the contribution of gravitational settling should be incorporated into the nanoparticle total mass flux and into the conservation of energy.²⁸

A. Governing equations and boundary conditions

The set of non-dimensional governing equations comprising the continuity, momentum conservation, advection-conduction heat transfer, and the advection-mass transfer is as follows:²⁸

$$P(\rho_{nf})_t + \nabla \cdot (\rho_{nf} \mathbf{u}) = 0, \quad (9a)$$

$$(\rho_{nf} \mathbf{u})_t + P^{-1} \nabla \cdot (\rho_{nf} \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} - \rho_{nf} \mathbf{G} \mathbf{e}_z, \quad (9b)$$

$$(\rho c)_{nf} [PT_t + (\mathbf{u} \cdot \nabla) T] = \nabla \cdot (K_{nf} \nabla T), \quad (9c)$$

$$PL^{-1} \phi_t + L^{-1} \nabla \cdot (\mathbf{u} \phi) = \nabla^2 \phi + \nabla \cdot (\eta \phi \nabla T), \quad (9d)$$

with $\nabla \equiv (\partial_x, \partial_z)$, $\mathbf{u} = (u, w)$, t is time, and $\boldsymbol{\tau}$ is the deviatoric part of the stress tensor, namely,

$$\boldsymbol{\tau} = \mu_{nf} (\nabla \mathbf{u} + \nabla \mathbf{u}^\top) - \frac{2}{3} \mu_{nf} (\nabla \cdot \mathbf{u}), \quad (10)$$

with the superscript $^\top$ denoting the transpose of the corresponding tensor. From here and on, subscripts x , z , and t denote partial differentiation with respect to the corresponding variable. We note that Stokes' hypothesis⁴⁹ of a zero value for the dilatational viscosity is used here.

The boundary conditions at the planar solid substrate, $z = 0$ are no-slip-no-penetration, a specified heat flux, and impermeability in terms of mass transfer, respectively,

$$\mathbf{u} = 0, \quad K_{nf} T_z = -1, \quad \phi_z + \eta \phi T_z = 0. \quad (11a)$$

The boundary conditions at the deformable interface, $z = h(x, z, t)$ are, respectively, the kinematic boundary condition, the balance of normal and tangential stresses, the continuity of the heat flux, and impermeability for mass transfer

$$Ph_t + uh_x = w, \quad (11b)$$

$$-p - \frac{2\mu_{nf}}{3(1+h_x^2)} [h_x(-2h_x u_x + 3w_x) + 3h_x u_z + (h_x^2 - 2)w_z + u_x] = \frac{\Sigma_0 h_{xx}}{(1+h_x^2)^{3/2}}, \quad (11c)$$

$$\mu_{nf} [2h_x(u_x - w_z) + (h_x^2 - 1)(u_z + w_x)] = [M_T(T_x + h_x T_z) + M_S(\phi_x + h_x \phi_z)] \sqrt{1+h_x^2}, \quad (11d)$$

$$K_{nf}(T_z - h_x T_x) + BT \sqrt{1+h_x^2} = 0, \quad (11e)$$

$$\phi_z - h_x \phi_x + \eta \phi (T_z - h_x T_x) = 0.$$

Finally, we have to satisfy the conservation of the average particle concentration in a layer by

$$\int_0^h \phi(x, z, t) dz = \Phi \quad (12)$$

for a specified positive value Φ .

The system variables are non-dimensionalized via

$$(x^*, z^*) = h_0^*(x, z), \quad h^*(x^*, z^*, t^*) = h_0^* h(x, z, t), \quad (13)$$

$$t^* = \frac{h_0^{*2}}{\nu_{bf}^*} t, \quad \mathbf{u}^* = \frac{\kappa_{bf}^*}{h_0^*} \mathbf{u}, \quad p^* = p_\infty^* + \frac{\mu_{bf}^* \kappa_{bf}^*}{h_0^{*2}} p,$$

$$T^* = T_\infty^* + \frac{q^* h_0^*}{K_{bf}^*} T, \quad \phi^* = \phi_m \phi, \quad K_{nf}^* = K_{bf}^* K_{nf},$$

$$\rho_{nf}^* = \rho_{bf}^* \rho_{nf}, \quad \mu_{nf}^* = \mu_{bf}^* \mu_{nf}, \quad (\rho^* c^*)_{nf} = (\rho^* c^*)_{bf} (\rho c)_{nf},$$

where $\mathbf{u} = (u, w)$ is the flow field, with the dimensional variables decorated with an asterisk, and the corresponding dimensionless variables denoted with the same symbol but without the asterisk.

The dynamics of the system is governed by the following nondimensional parameters

$$P = \frac{\nu_{bf}^* \rho_{bf}^* c_{bf}^*}{K_{bf}^*}, \quad G = \frac{g^* h_0^{*3}}{\nu_{bf}^* \kappa_{bf}^*}, \quad L = \frac{C_B T_\infty^* \rho_{bf}^* c_{bf}^*}{K_{bf}^*},$$

$$B = \frac{\hat{q}^* h_0^*}{K_{bf}^*}, \quad M_T = \frac{q^* h_0^{*2} \sigma_{T^*}}{\mu_{bf}^* K_{bf}^* \kappa_{bf}^*}, \quad M_S = \frac{\phi_m h_0^* \sigma_{\phi^*}}{\mu_{bf}^* \kappa_{bf}^*}, \quad (14a)$$

$$\Sigma_0 = \frac{\sigma_r^* h_0^*}{\mu_{bf}^* \kappa_{bf}^*}, \quad \eta = \frac{C_T q^* h_0^*}{C_B T_\infty^{*2} K_{bf}^*}.$$

where P , G , L , B , M_T , M_S , Σ_0 , and η are, respectively, the Prandtl, the modified Galileo, Lewis, Biot, thermal Marangoni, solutal Marangoni, dimensionless surface tension (akin to the inverse capillary number) numbers, and the Soret coefficient. The nomenclature and typical values of the physical properties and of the dimensionless numbers used in this investigation are collected for the reader's

convenience in Table I. In addition, the dimensionless form of the thermophysical properties appearing in the set of Eqs. (9) and (11) are

$$\rho_{nf}(\phi) = (1 - \phi \phi_m) + \phi \phi_m \rho_{np},$$

$$(\rho c)_{nf}(\phi) = (1 - \phi \phi_m) + \phi \phi_m (\rho c)_{np}, \quad (15)$$

$$K_{nf}(\phi) = 1 + a \phi \phi_m, \quad \mu_{nf}(\phi) = (1 - \phi)^{-2}.$$

Note that in the definition of the Biot number B in Eq. (14a), $\hat{q}^*$ represents the heat transfer rate from the liquid phase to the gas phase by convection according to Newton's law of cooling.

B. Base state

We now determine a quiescent ($\mathbf{u} \equiv \mathbf{u}_0 = \mathbf{0}$) steady base state of the system with the flat interface $h = 1$ and with the temperature and concentration fields depending solely on z , $T = T_0(z)$, $\phi = \phi_0(z)$, of the problem given by Eqs. (9)–(11). It is determined by

$$p_{0,z} = -((1 - \phi_0 \phi_m) + \phi_0 \phi_m \rho_{np}) G, \quad (16a)$$

$$[(1 + a \phi_m \phi_0) T_{0,z}]_z = 0, \quad (16b)$$

$$(\phi_{0,z} + \eta \phi_0 T_{0,z})_z = 0, \quad (16c)$$

with boundary conditions

$$z = 0: \quad (1 + a \phi_m \phi_0) T_{0,z} = -1, \quad \phi_{0,z} + \eta \phi_0 T_{0,z} = 0, \quad (16d)$$

$$z = 1: \quad p_0 = 0, \quad (1 + a \phi_m \phi_0) T_{0,z} + B T_0 = 0, \quad (16e)$$

$$\phi_{0,z} + \eta \phi_0 T_{0,z} = 0.$$

Finally, following Eq. (12), we obtain

$$\int_0^1 \phi_0(z) dz = \Phi. \quad (16f)$$

The solution of the problem given by Eqs. (16) for the pressure, temperature, and particle concentration components is found to be

$$p_0 = G(1 - z) + \frac{1}{2\eta a \phi_m} (G \phi_m (\rho_p - 1) \times [\mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta)) - \mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta z))] \times [\mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta)) + \mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta z)) + 2]), \quad (17a)$$

$$T_0 = \frac{1}{\eta B} \left[B \ln \left(\mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta)) \right) - B \ln \left(\mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta z)) \right) + \eta \right], \quad (17b)$$

$$\phi_0 = \frac{1}{a \phi_m} \mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta z)), \quad (17c)$$

where $\mathcal{W}(z)$ is the principal (increasing) branch of the Lambert W function⁵⁰ defined via

TABLE I. Nomenclature and typical values of the physical properties and dimensionless numbers used here.

Parameter	Symbol	Values
Equilibrium nanofluid layer thickness	h_0^*	$\in (10^{-6}, 10^{-4})$ m
Gravity	g^*	$\in (0.981, 9.81)$ m/s ²
Base fluid viscosity	μ_{bf}^*	10^{-3} kg/(m s)
Base fluid density	ρ_{bf}^*	1000 kg/m ³
Base fluid kinematic viscosity	ν_{bf}^*	10^{-6} m ² /s
Base fluid thermal conductivity	K_{bf}^*	0.61 W/(m · K)
Base fluid heat capacity	c_{bf}^*	≈ 4.18 kJ/(kg K)
Base fluid thermal diffusivity	κ_{bf}^*	0.146×10^{-6} m ² /s
Ambient gas temperature	T_∞^*	≈ 300 K
Equilibrium surface tension	σ_r^*	$\in (10^{-3}, 30. \times 10^{-3})$ N/m
Temperature gradient of the surface tension	$\sigma_{T^*}^*$	0.069×10^{-3} N/(m K)
Concentration gradient of the surface tension	$\sigma_{\phi^*}^*$	$\in (0.05 \times 10^{-3}, 5. \times 10^{-3})$ N/m
Temperature difference across the nanofluid layer	ΔT^*	$\in (0.1, 10.)$ K
Random close-packing volume fraction (RCP)	ϕ_m	≈ 0.65
Nanoparticle average bulk concentration	Φ	$\in (0.01, 0.027)$
Nanoparticle diameter	d_p^*	≈ 10 nm
Boltzmann constant	K_B	$1.380\,649 \times 10^{-23}$ J/K
Nanoparticle density	ρ_{np}^*	~ 4000 kg/m ³
Nanoparticle heat capacity	c_{np}^*	≈ 0.718 kJ/(kg K)
Prandtl number	P	7
Modified Galileo number	G	$\in (10^{-4}, 67.)$
Dimensionless surface tension	Σ_0	$\in (1., 2. \times 10^4)$
Biot number	B	$\in (10^{-8}, 10^{-1})$
Lewis number	L	$\in (10^{-3}, 4.22 \times 10^{-4})$
Soret coefficient	η	$\in (0.31, 3.1)$

$$z = \mathcal{W}(z) \exp \mathcal{W}(z) \quad (18)$$

and $\gamma(\Phi)$ is determined from the substitution of $\phi_0(z)$ from Eq. (17c) into Eq. (16f) (a thorough derivation of the base state is provided in Appendix A of Gandhi *et al.*²⁸).

The left and right panels of Fig. 2 represent the base state temperature $T_0(z)$ and concentration $\phi_0(z)$ at different values of the average bulk nanoparticle concentration Φ and the temperature drop across the layer in the base state ΔT^* determined via

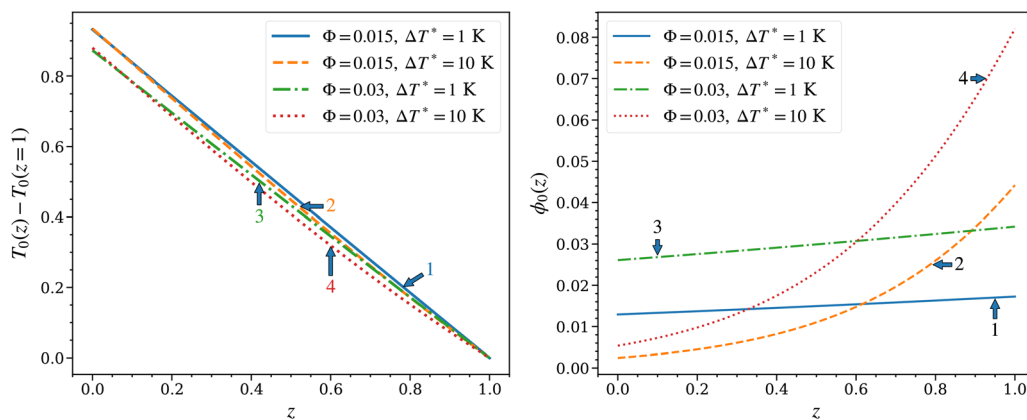


FIG. 2. Variation of the quiescent base-state temperature presented as $T_0(z) - T_0(z=1)$ (left panel) and concentration $\phi_0(z)$ (right panel) with height z for different values of the average bulk concentration Φ and the temperature difference ΔT^* with $B = 0.01$, $\eta = 0.31$, and $a = 7.47$. Curves 1 and 3 represent the base state of the system with the average bulk nanoparticle concentration $\Phi = 0.015$ and $\Phi = 0.03$ at $\Delta T^* = 1$ K, respectively. Curves 2 and 4 depict the base state of the system with the average bulk nanoparticle concentration $\Phi = 0.015$ and $\Phi = 0.03$ at $\Delta T^* = 10$ K, respectively.

$$\Delta T^* = q^* h_0^* / K_{bf}^*. \quad (19)$$

We identify that the concentration profiles exhibit an unstable stratification, owing to the positive value of the Soret coefficient η . Interestingly, we observe that the concentration profiles display a non-linear variation across the layer thickness z for higher values of the temperature difference ΔT^* , e.g., $\Delta T^* = 10$ K. The influence of nonlinearity becomes more pronounced for high Φ due to the concentration-dependent Soret effect.

III. LINEAR STABILITY ANALYSIS

We now investigate the linear stability of the base state given by Eq. (17) by representing the dependent variables around the base state in the form

$$\begin{aligned} \mathbf{u} &= \mathbf{u}_0 + \bar{\mathbf{u}} = \bar{\mathbf{u}}, \quad p = p_0 + \bar{p}, \quad T = T_0 + \bar{T}, \\ \phi &= \phi_0 + \bar{\phi}, \quad h = h_0 + \bar{\zeta}, \end{aligned} \quad (20)$$

where the values with an overbar depict the disturbances of the corresponding fields. We also linearize the thermophysical properties around the corresponding values related to the dimensionless base state concentration ϕ_0 , i.e.,

$$\begin{aligned} \rho_{nf} &= R_0 + \bar{R} = (\phi_m(\rho_{np} - 1)\phi_0 + 1) + (\phi_m(\rho_{np} - 1)\bar{\phi}), \\ (\rho c)_{nf} &= H_0 + \bar{H} = (\phi_m((\rho c)_{np} - 1)\phi_0 + 1) \\ &\quad + (\phi_m((\rho c)_{np} - 1)\bar{\phi}), \\ K_{nf} &= K_0 + \bar{K} = (a\phi_m\phi_0 + 1) + \bar{\phi}a\phi_m, \\ \mu_{nf} &= M_0 + \bar{M} = \frac{1}{(\phi_0 - 1)^2} - \frac{2\bar{\phi}}{(\phi_0 - 1)^3}, \end{aligned} \quad (21)$$

where, for brevity, the symbols $\bar{X} \equiv (\bar{R}, \bar{M}, \bar{K}, \bar{H})$ stand for disturbances of ρ_{nf} , μ_{nf} , K_{nf} , and $(\rho c)_{nf}$, respectively, represented each by the second term in the respective equation in Eq. (21). The values of R_0 , M_0 , K_0 , and H_0 are given by the first term in the corresponding equation in Eq. (21).

We rewrite the two-dimensional linearized equations in terms of the vertical velocity disturbance $\bar{w}$ by satisfying the continuity equation. In what follows, we apply the $\nabla \times$ operator to the momentum conservation equations, and differentiate the normal stress balance boundary condition along the interface, followed by substituting the pressure gradient components obtained from the x, z -components of the momentum conservation equations, thereby completely eliminating the pressure from the equations.

We introduce the normal perturbation modes in the form

$$\begin{aligned} (\bar{u}, \bar{w}, \bar{T}, \bar{\phi}, \bar{\zeta}; \bar{X})(x, z, t) &= (u(z), w(z), \theta(z), \\ &\quad \varphi(z), \zeta; X) \exp(ikx + \lambda t), \end{aligned} \quad (22)$$

where $u, w, \theta, \varphi, \zeta$, and $X \equiv (R(z), M(z), K(z), H(z))$ are the amplitudes of the horizontal and vertical fluid velocity components, temperature, concentration, interfacial deformation, and perturbations of the thermophysical properties (expressed via φ), respectively. We denote $k \in \mathbb{R}$ and $\lambda \in \mathbb{C}$ as the wavenumber and the growth rate of the disturbances, respectively. In this representation, positive and

negative values of the real part of λ correspond to instability and stability regimes of the base state, respectively.

The eigenvalue problem (EVP), which determines the growth rate λ of the disturbance as a function of the wavenumber k and the problem parameters reads

$$R_0(w' + iku) + \lambda PR + wR_0' = 0, \quad (23a)$$

$$\begin{aligned} -iGkR + k^2M_0u' + 2k^2uM_0' - ikw(k^2M_0 + \lambda R_0 + M_0') \\ + ikM_0w'' - u'''M_0 - 2u''M_0' + \lambda R_0u' - u'M_0'' + \lambda uR_0' = 0, \end{aligned} \quad (23b)$$

$$-T_0''K + \theta(k^2K_0 + \lambda PH_0) + T_0'(wH_0 - K') - K_0\theta'' - K_0'\theta' = 0, \quad (23c)$$

$$\begin{aligned} \varphi(k^2L - \eta LT_0'' + \lambda P) + \eta k^2L\theta\phi_0 + \phi_0(iku - \eta L\theta'' + w') \\ + \phi_0'(w - \eta L\theta') - \eta LT_0'\phi' - L\varphi'' = 0. \end{aligned} \quad (23d)$$

The boundary conditions at $z = 0$ are

$$\begin{aligned} \mathbf{u} = 0, \quad \theta'(a\phi_m\phi_0 + 1) + a\phi_m\varphi T_0' = 0, \\ \eta\phi_0\theta' + \eta\varphi T_0' + \varphi' = 0. \end{aligned} \quad (23e)$$

The boundary conditions at the deformable interface $z = h$ are

$$\zeta\lambda P - w = 0, \quad (23f)$$

$$\begin{aligned} T_0'(a\phi_m(\zeta\phi_0' + \varphi) + B\zeta) + (a\phi_m\phi_0 + 1)\theta' \\ + \zeta T_0'(a\phi_m\phi_0 + 1) + B\theta = 0, \end{aligned} \quad (23g)$$

$$\zeta\phi_0'' + \eta\phi_0(\theta' + \zeta T_0'') + \eta T_0'(\zeta\phi_0' + \varphi) + \varphi' = 0, \quad (23h)$$

$$kM_S(\zeta\phi_0' + \varphi) + kM_T(\zeta T_0' + \theta) + M_0(kw - iu') = 0, \quad (23i)$$

$$\begin{aligned} i\zeta k^3\Sigma_0 + u(2k^2M_0 + \lambda R_0) + i(-\zeta kp_0' + kM_0w' - kM_0'u) \\ + iM_0u'' + iM_0'u' = 0. \end{aligned} \quad (23j)$$

Here, thermophysical properties bearing subscripts with and without zero represent the material properties with the steady state nanoparticle concentration and their perturbations due to perturbed nanoparticle concentration, respectively [see Eq. (21)]. We note that the normal perturbation of the concentration field $\bar{\phi}$ introduces a disturbance of the nanoparticle concentration, and consequently, a disturbance of the temperature through the thermophysical properties of the fluid across the nanofluid layer. We also note that the average nanoparticle concentration for normal disturbances is automatically set to zero, since the corresponding integral vanishes for any periodic domain.

A comment on the perturbed heat flux boundary condition at the substrate (23e) is in order. The case of a specified heat flux at the solid substrate may be considered as a particular case of the more general one in which the substrate, subjected to heat transfer at its bottom, is thick. The boundary condition at the solid-liquid interface would arise from the continuity of both the temperature and the conductive heat flux. The latter would have been written in the perturbed state at every point of the solid-liquid interface as^{51–53}

$$\mathcal{Q}_{nf} = \mathcal{Q}_s, \quad (24a)$$

where

$$\begin{aligned}\mathcal{Q}_{nf} &= 1 + \frac{\delta q_{nf}^*}{q^*} = 1 + \delta(-K_{nf}(\phi)T') \\ &= 1 + \delta(-K_{nf}(\phi))T'_0 - K_{nf}(\phi_0)\delta T',\end{aligned}\quad (24b)$$

$$\mathcal{Q}_s = 1 + \frac{\delta q_s^*}{q^*} = 1 - K_s\delta T'_s, \quad (24c)$$

represent the fluxes in the nanofluid and solid, respectively, the subscripts nf , s denote the values related to the nanofluid and the solid, respectively, q^* is the specified constant heat flux imposed at the solid substrate at every point along the bottom, and δX represents the disturbance of the quantity X . Equating the fluxes (24b) and (24c) in the solid and fluid layers, we obtain at every point along the solid-liquid boundary

$$\frac{\delta(K_{nf}(\phi))T'_0}{K_{nf}(\phi_0)} + \delta T' = \bar{K}\delta T'_s, \quad (25)$$

where $\bar{K} = \frac{K_s}{K_{nf}(\phi_0)}$ is the dimensionless ratio of the thermal conductivity of the solid substrate and that of the nanofluid at the particle concentration in the base state. We note that in the particular case of a poorly conducting substrate $K_s \ll K_{nf}$, equivalent to $\bar{K} \ll 1$, the right-hand side of Eq. (25) becomes negligible, and in the case of an infinitesimally thin substrate, the boundary condition (23e) follows at every point along the solid-liquid interface. It is important to note that the particle concentration-dependent thermal conductivity of the nanofluid induces in the perturbed state the emergence of two components of the heat flux, namely, due to disturbances of the substrate temperature and powered by convection resulting from the presence of the imposed heat flux⁵² as evident from Eq. (25). We reiterate that the prescribed heat flux boundary condition used here and the ensuing boundary condition (23e) in the perturbed state at the solid-liquid interface are appropriate in the case of a *poorly conducting* substrate.

Before we embark on the long-wave investigation, we rewrite the non-dimensional form of the nanofluid continuity Eq. (9a) as

$$\nabla \cdot \mathbf{u} = -(\rho_{np} - 1)\phi_m[P\phi_t + \nabla \cdot (\phi\mathbf{u})]. \quad (26)$$

Based on Eq. (26) and on the nanoparticle mass flux balance Eq. (9d), the divergence of the velocity vector is recast in the form

$$\nabla \cdot \mathbf{u} = -L(\rho_{np} - 1)\phi_m[\nabla^2\phi + \nabla \cdot (\eta\phi\nabla T)]. \quad (27)$$

In the case of a moderately dense nanoparticle suspension $\Phi \leq 0.1$ and a small Lewis number $L \in (10^{-4}, 10^{-2})$, the right-hand side of Eq. (27) subsequently vanishes. Hence, this signifies that the moderately dense nanoparticle suspension can be treated as incompressible by neglecting the right-hand side of Eq. (26). In our previous studies,^{28,29} we validated the underlying simplification by direct comparison between the results arising from the EVP (23) in its compressible and incompressible versions.

IV. LONG-WAVE EXPANSION

We first recall that the shape of the neutral curve $M(k)$ for a long-wave deformational Marangoni instability is determined by the Biot number B and the non-dimensional surface tension parameter Σ_0 in the form

$$M(k^2) = M_0 + c_1 B/k^2 + c_2 \Sigma_0 k^2, \quad (28a)$$

where c_1 and c_2 are constants.^{54,55} Based on this expression, one finds the scale for the critical wavenumber which yields the balance between the two terms mentioned above, namely,

$$k_c \sim (B/\Sigma_0)^{1/4}. \quad (28b)$$

The following two possible scale selections allow the long-wave instability $k \sim \varepsilon$ may be deduced: (i) $\Sigma_0 = O(1)$ and $B = O(\varepsilon^4)$, and (ii) $\Sigma_0 = O(\varepsilon^{-2})$ and $B = O(\varepsilon^2)$. For both cases, Eq. (28b) yields $k_c = O(\varepsilon)$. As will be seen below, in case (i), we obtain the terms containing Σ_0 and $\beta = B\varepsilon^{-4}$ in the solvability condition at a higher order of expansion in powers of ε . However, in case (ii), we obtain the essential contributions of both the Biot number and surface tension already at the second order of expansion.

This first scaling law is quite typical for dilute binary mixtures. However, it was also shown that for dilute binary mixtures, the onset of instability may take place for even smaller wavenumbers for $B \ll 1$, so scaling $k_c \sim B^{1/2}$ may also be relevant^{19,56} here.

In what follows, we introduce the following scaling with a small parameter $\varepsilon \ll 1$ into the eigenvalue problem (23):

$$B = \varepsilon^2\beta, \quad k = \varepsilon K, \quad \lambda = \varepsilon^2\Lambda, \quad \Sigma_0 = \varepsilon^{-2}\Sigma_s. \quad (29)$$

We expand the perturbed variables w , ϕ , θ , ζ , the growth rate λ , and the thermal and solutal Marangoni numbers M_T and M_S , respectively, into the series of ε via

$$\begin{aligned}w &= \varepsilon^2(w_0 + \varepsilon^2 w_2 + \dots), \quad (\theta, \phi, \zeta) \\ &= (\theta_0, \phi_0, \zeta_0) + \varepsilon^2(\theta_2, \phi_2, \zeta_2) + \dots, \\ (\Lambda, M_T, M_S) &= (\Lambda_0, M_{T0}, M_{S0}) + \varepsilon^2(\Lambda_2, M_{T2}, M_{S2}) + \dots.\end{aligned}\quad (30)$$

The substitution of the asymptotic expansions (29) and (30) into the eigenvalue problem (23) results in a hierarchy of linear boundary value problems at different orders of ε . We note that, at leading order, the linear boundary-value problem is identical (except for the boundary condition for the interfacial deformation) to that presented by Gandhi *et al.*²⁹ For the convenience of the reader, we present the details of the leading-order problem, the corresponding solutions, and the adjoint problem with its solution in Appendix.

The energy and nanoparticle concentration equations with corresponding boundary conditions at $O(\varepsilon^2)$ read

$$\begin{aligned}& -\frac{\eta\mathcal{P}(a\phi_m\phi_2 + (\mathcal{P}+1)^2\mathcal{D}\theta_2)}{(\mathcal{P}+1)^3} + \frac{a\phi_m\mathcal{D}\phi_2}{\mathcal{P}+1} - (\mathcal{P}+1)\mathcal{D}^2\theta_2 \\ &= -\frac{\theta_0[a(K^2(\mathcal{P}+1) + \Lambda_0P) + \Lambda_0P(\rho_{cnp} - 1)\mathcal{P}]}{a} \\ &+ \frac{[a + (\rho_{cnp} - 1)\mathcal{P}]w_0}{a(\mathcal{P}+1)},\end{aligned}\quad (31a)$$

$$\begin{aligned}& -\frac{\eta\mathcal{P}(\eta\mathcal{D}\theta_2 + (\mathcal{P}+1)\mathcal{D}^2\theta_2)}{a\phi_m(\mathcal{P}+1)} + \frac{\eta[(\mathcal{P}+1)^2\mathcal{D}\phi_2 - \eta\mathcal{P}\phi_2]}{(\mathcal{P}+1)^3} - \mathcal{D}^2\phi_2 \\ &= -\frac{\eta K^2\mathcal{P}\theta_0}{a\phi_m} - \frac{\eta\mathcal{P}w_0}{La\phi_m(\mathcal{P}+1)} - \frac{\phi_0(K^2L + \Lambda_0P)}{L},\end{aligned}\quad (31b)$$

$$(\mathcal{P}(0) + 1)\mathcal{D}\theta_2 - \frac{a\phi_m\phi_2}{\mathcal{P}(0) + 1} = 0,$$

$$\frac{\eta\mathcal{P}(0)\mathcal{D}\theta_2}{a\phi_m} - \frac{\eta\phi_2}{\mathcal{P}(0) + 1} + \mathcal{D}\phi_2 = 0 \quad \text{at } z = 0, \quad (31c)$$

$$\begin{aligned}
 (\mathcal{P}(1) + 1)\mathcal{D}\theta_2 - \frac{\beta\zeta_0}{\mathcal{P}(1) + 1} &= \frac{a\phi_m\varphi_2}{\mathcal{P}(1) + 1} - \beta\theta_0, \\
 \frac{\eta\mathcal{P}(1)\mathcal{D}\theta_2}{a\phi_m} - \frac{\eta\varphi_2}{\mathcal{P}(1) + 1} + \mathcal{D}\varphi_2 &= 0 \quad \text{at } z = 1.
 \end{aligned} \tag{31d}$$

We now rewrite the second-order Eqs. (31a) and (31b) in the matrix form

$$\mathcal{L}\mathbf{A}_2 = \mathbf{R}, \tag{32a}$$

$$\mathbf{R} = - \begin{pmatrix} \frac{\theta_0[a(K^2(\mathcal{P} + 1) + \Lambda_0 P) + \Lambda_0 P(\rho c_{np} - 1)\mathcal{P}]}{a} - \frac{[a + (\rho c_{np} - 1)\mathcal{P}]w_0}{a(\mathcal{P} + 1)} + \beta\theta_0(1) - \frac{\beta\zeta_0(1)}{\mathcal{P}(1) + 1} \\ \frac{\eta K^2 \mathcal{P}\theta_0}{a\phi_m} + \frac{\eta \mathcal{P}w_0}{La\phi_m(\mathcal{P} + 1)} + \frac{\varphi_0(K^2 L + \Lambda_0 P)}{L} \end{pmatrix}, \tag{32b}$$

where $\mathbf{A}_2 = \begin{pmatrix} \theta_2 \\ \varphi_2 \end{pmatrix}$.

The sets of linear equations (A1c) and (32a) possess the same matrix $\mathcal{L}$, while the former is homogeneous and the latter is nonhomogeneous. Therefore, we use the Fredholm alternative theorem^{17,22,55,57–60} to impose the solvability condition that, in the case at hand, consists of two conditions of orthogonality between the vector $\mathbf{R}$ and the eigenvectors of the adjoint operator $\mathbf{u}_1^*$ and $\mathbf{u}_2^*$

$$\langle \mathbf{R}, \mathbf{u}_1^* \rangle = 0 \quad \text{and} \quad \langle \mathbf{R}, \mathbf{u}_2^* \rangle = 0. \tag{33}$$

Due to the simple form of $\mathbf{u}_1^*$ and $\mathbf{u}_2^*$ in Eq. (A8), both equations in Eq. (33) represent just separate integrations of the components of the vector $\mathbf{R}$ across the layer with their results equated to zero.

The problem (A9) is solved numerically in terms of w_0 with $\begin{pmatrix} \theta_0 \\ \varphi_0 \end{pmatrix}$ taken alternately as $\mathbf{u}_1$ and $\mathbf{u}_2$ with the parameters M_{T0} and M_{S0} for fixed G , η , a , ϕ_m , ρ_{np} , Φ , and Σ_s , resulting in the solutions $w_0 = w_0^{(1)}(M_{T0}, M_{S0}, \Lambda_0)$ and $w_0 = w_0^{(2)}(M_{T0}, M_{S0}, \Lambda_0)$, respectively. Thus, we express the general solution for $w_0(M_{T0}, M_{S0}, \Lambda_0)$ as a linear combination of these, namely $w_0(M_{T0}, M_{S0}, \Lambda_0) = \alpha_1 w_0^{(1)}(M_{T0}, M_{S0}, \Lambda_0) + \alpha_2 w_0^{(2)}(M_{T0}, M_{S0}, \Lambda_0)$.

We rewrite the solvability conditions (33) as

$$\begin{aligned}
 \int_0^1 \left[\frac{\theta_0[a(K^2(\mathcal{P} + 1) + \Lambda_0 P) + \Lambda_0 P(\rho c_{np} - 1)\mathcal{P}]}{a} \right. \\
 \left. - \frac{[a + \mathcal{P}(\rho c_{np} - 1)]w_0}{(\mathcal{P} + 1)a} \right] dz \\
 + \left(\beta\theta_0(1) - \frac{\beta\zeta_0(1)}{\mathcal{P}(1) + 1} \right) = 0,
 \end{aligned} \tag{34}$$

$$\int_0^1 \left[\frac{\eta K^2 \mathcal{P}\theta_0}{a\phi_m} + \frac{\eta \mathcal{P}w_0}{La\phi_m(\mathcal{P} + 1)} + \frac{\varphi_0(K^2 L + \Lambda_0 P)}{L} \right] dz = 0,$$

with θ_0 and φ_0 given from Eq. (A4). Substituting θ_0 , φ_0 , and $w_0(M_{T0}, M_{S0}, \Lambda_0)$ into Eq. (34) results in a linear set of two homogeneous equations with respect to the unknowns α_1 and α_2 . Since $w_0(M_{T0}, M_{S0}, \Lambda_0)$ depends linearly on both M_{T0} and M_{S0} via the boundary condition (A9d), all four entries of the resulting matrix of this set of equations are linear functions of M_{T0} and M_{S0} . We obtain

the critical values of the threshold values M_{T0} by fixing the set of parameters, including M_{S0} , and imposing the condition of a zero value to the matrix determinant, resulting in a quadratic equation in terms of M_{T0} . The growth rate Λ_0 is determined by substitution of the values of the thermal and solutal Marangoni numbers along the critical curve into the determinant of the solvability condition and finding the value of Λ_0 , which provides us with the dispersion relation for the thermosolutal Marangoni instability. The real part of the growth rate $\Lambda_{0r} \equiv \Re(\Lambda_0)$ determines the stability of the system, whereas its imaginary component $\Lambda_{0i} \equiv \Im(\Lambda_0)$ yields the critical frequency for the oscillatory instability.

V. RESULTS AND DISCUSSION

The emergence of both monotonic and oscillatory thermosolutal instabilities in a nanofluid layer with a deformable interface heated at the substrate underneath the layer was recently reported.²⁸ We also showed²⁹ that in the limit of small Biot numbers and a large surface tension number, equivalent to a weak interfacial deformability, the onset of the thermosolutocapillary monotonic instability may be adequately determined using the long-wave asymptotic expansions. However, the role of interfacial deformation is very important in the long-wave limit and needs to be further investigated.

Figure 3 presents an excellent agreement between the results obtained using the long-wave approach developed in Sec. IV and those of the numerical solution of the incompressible version of the full eigenvalue problem (23) using the approach presented in Ref. 28. The left panel of Fig. 3 displays the variation of the critical wavenumber k_c for the oscillatory instability with the dimensionless inverse surface tension number Σ_0 . We observe that the critical wavenumber k_c follows the power-law decrease $k_c \sim \Sigma_0^{-1/4}$ with Σ_0 , while the inset of the left panel of Fig. 3 illustrates the power-law decrease in the critical frequency $\lambda_i = \Im(\lambda) \sim \Sigma_0^{-1/2}$ of the oscillatory instability with Σ_0 . In turn, this result supports the fact that the surface tension effectively reduces the interfacial deformation or the distance between the substrate and the troughs and damps the presence of the oscillations generated by thermocapillarity. As a result, emerging cells extend farther, tending to decrease the critical wavenumber.^{61,62} The right panel of Fig. 3 presents a linear increase in the critical thermocapillary Marangoni number M_T with $\Sigma_0^{1/2}$; therefore, we note that the surface tension represents a stabilizing effect via Σ_0 . The dot-dashed line in the right panel of Fig. 3 shows the linear fit $M_T = M_0 + c_1 \Sigma_0^{1/2}$, where the constant c_1 contains the contributions of other parameters of the

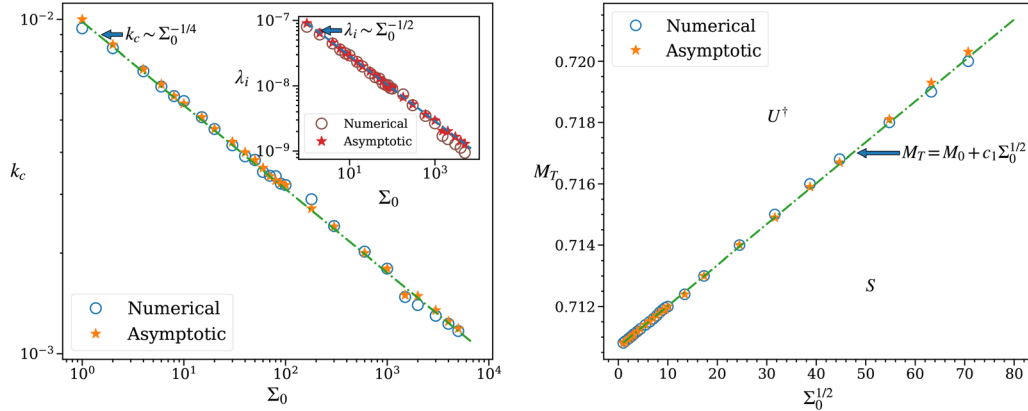


FIG. 3. Pure thermocapillary instability with $\Phi = 0.01$, $\eta = 0.31$, $G = 1$, $M_S = 0$, $a = 7.47$, $L = 10^{-3}$, and $B = 10^{-8}$. Left panel: Variation of the critical wavenumber k_c with the dimensionless surface tension number Σ_0 ; Inset: Variation of the critical frequency $\lambda_i = \Im(\lambda)$ with Σ_0 . Right panel: Variation of the critical value of M_T with $\Sigma_0^{1/2}$. Here, the $\circ$ and $\star$ points denote the critical values found numerically by solving the incompressible version²⁸ of the EVP (23) and by the long-wave asymptotic expansion of Sec. IV, respectively. The dot-dashed and dashed lines in the left panel and in its inset represent the data fits for $k_c \sim \Sigma_0^{-1/4}$ and $\lambda_i \sim \Sigma_0^{-1/2}$ with the proportionality coefficients 9.84×10^{-3} and 9.11×10^{-8} , respectively. The dot-dashed line in the right panel shows the linear data fit $M_T = M_0 + c_1 \Sigma_0^{1/2}$ with $M_0 = 0.71$ and $c_1 = 1.33 \times 10^{-4}$. The symbols $U^\dagger$ and S stand for the domains of oscillatory thermocapillary instability and system stability, respectively.

system. It is noteworthy to point out that Eq. (28a) is well balanced at criticality, since $M_T - M_0$ and $\Sigma_0 k_c^2$ are both proportional to $\Sigma_0^{1/2}$ at the onset of oscillatory instability considered here.

We also find that the critical wavenumber of the oscillatory instability follows the standard long-wave scaling^{17,19} with the Biot number B , namely $k_c \sim B^{1/4}$, as shown in the left panel of Fig. 4. The inset shows a slow increase in the oscillation frequency $\lambda_i \equiv \Im(\lambda) \sim B^{1/2}$ with B . An increase in the Biot number leads to heat transfer to the environment at the layer interface; therefore, the system is stabilized as follows from an increase in the critical Marangoni number M_T shown in the right panel of Fig. 4.

Interestingly, we note that M_T linearly varies with $B^{1/2}$ as well, as expressed by an excellent agreement between the results obtained from

the asymptotic analysis of Sec. IV and the data fit $M_T = M_0 + c_2 B^{1/2}$ shown in Fig. 4. Here, the reference value of the Marangoni number M_0 is identical to that obtained for its variation with Σ_0 and the constant c_2 that contains contributions from other parameters. We emphasize that the emergence of oscillatory thermocapillary instability is due to a collective action of gravity, thermal conductivity of the nanofluid, and the slow diffusion rate of nanoparticles.

We recall that with a positive Soret coefficient η , as in this paper, the base-state concentration of nanoparticle exhibits an unstable stratification, as shown in Fig. 2. Thus, solutocapillarity acts as a stabilizing mechanism. Furthermore, Gandhi *et al.*²⁸ found that for a sufficiently thin layer or for low values of the modified Galileo number G in a layer with a deformable interface, the monotonic thermocapillary instability

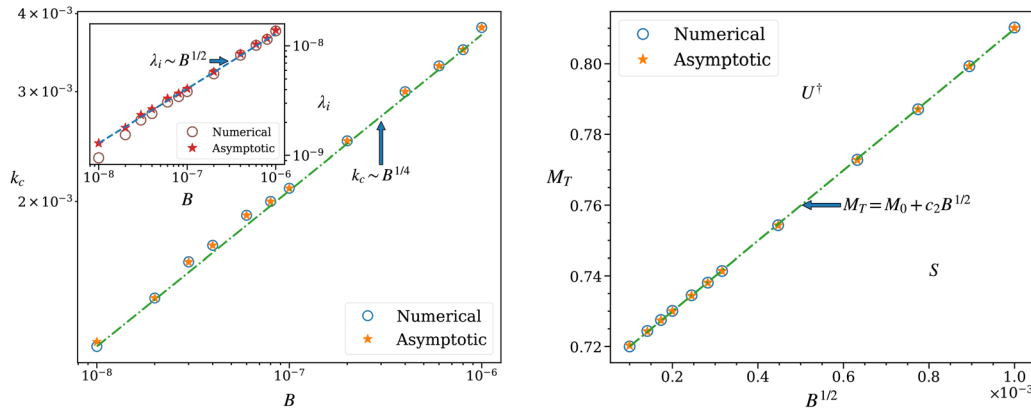


FIG. 4. Pure thermocapillary instability with $\Phi = 0.01$, $\eta = 0.31$, $G = 1$, $M_S = 0$, $a = 7.47$, $L = 10^{-3}$, and $\Sigma_0 = 5 \times 10^3$. The $\circ$ and $\star$ points denote the critical values found numerically by solving the incompressible version of the EVP (23) and by the long-wave asymptotic expansion of Sec. IV, respectively. Left panel: Variation of the critical value of the wave number k_c with the Biot number B ; Inset: Variation of the critical frequency $\lambda_i = \Im(\lambda)$ with B . The dot-dashed and dashed lines in the left panel and in the inset represent the data fits $k_c \sim B^{1/4}$ and $\lambda_i \sim B^{1/2}$ with the proportionality coefficients 0.117 and 1.29×10^{-5} , respectively. Right panel: Variation of the critical value of the thermal Marangoni number M_T with $B^{1/2}$. The dot-dashed line in the right panel denotes the linear data fit $M_T = M_0 + c_2 B^{1/2}$ with $M_0 = 0.71$ and $c_2 = 99.7578$. The symbols $U^\dagger$ and S stand for the domains of oscillatory thermocapillary instability and system stability, respectively.

prevails. In what follows, we study the onset of the combined long-wave thermo-solutocapillary instability.

The left panel of Fig. 5 illustrates the variation of the critical thermocapillary Marangoni number M_{T0} with the solutocapillary Marangoni number M_{S0} . We observe that for relatively small values of M_{S0} , the monotonic instability dominates. However, the oscillatory instability emerges with an increase in M_{S0} , for $M_{S0} \geq 2.8$, where the oscillatory branch bifurcates off the monotonic branch. It is interesting to note from the right panel of Fig. 5 that the critical wavenumber K_c for the monotonic instability remains almost constant with M_{S0} , whereas the critical wavenumber for the oscillatory instability changes via a discontinuity, increasing by some 50% with respect to its value corresponding to the monotonic branch. This implies that the oscillatory instability emerges via a codimension-two point, where the monotonic and oscillatory thresholds have the same critical value of the thermal Marangoni number M_{T0} and the competition between these occurs.^{28,63} We also note that the critical frequency $\Lambda_{0i} = \Im(\Lambda_0)$ for the oscillatory instability increases linearly with M_{S0} , as shown in the inset in the right panel of Fig. 5. We infer that the origin of the oscillatory instability observed in the $M_{S0} - M_{T0}$ plane is due to the disparity between the thermal and nanoparticle concentration diffusion time scales, expressed by the Lewis number L , with $L \ll 1$.^{17,18,28}

We now address the effect of the variation of the thermal conductivity stratification parameter a on the critical values of the long-wave thermocapillary instability. In our previous work,²⁸ we found that the combined action of particle diffusion and gravity expressed via interfacial deformation induces the emergence of oscillatory thermocapillary instability, where the former contributes significantly more than the latter. The left panel of Fig. 6 illustrates the effect of the thermal conductivity stratification parameter a on the threshold of the oscillatory thermocapillary instability. We note that thermal conductivity stratification stabilizes the long-wave oscillatory thermocapillary instability, which is expressed by an increase in M_T with a . However, we find that for low values of a , that is, when the thermal conductivity of the nanofluid depends weakly on the particle concentration, with a decrease in

the average bulk nanoparticle concentration Φ the long-wave thermocapillary instability exhibits a transition from the oscillatory instability to the monotonic instability, as shown in the right panel of Fig. 6. We note that in the limit of the vanishing Soret effect $\eta \ll 1$, the critical Marangoni number for the monotonic instability reduces to the analytical expression

$$M_T = \frac{48G(K^2 + \beta)}{(72 + G)K^2}, \quad (35)$$

given in Podolny *et al.*,¹⁹ where the mass flux driven by the Soret effect was assumed to be proportional to the temperature gradient with the factor χ there in the limit of $\chi \rightarrow 0$. We observe that the critical Marangoni number for the monotonic thermocapillary instability is slightly off the known value of $\frac{2}{3}G$ obtained for a layer of a pure Newtonian liquid.⁹ It is easily seen that the contribution of a non-zero wavenumber K lowers the value of M_T in Eq. (35). This is illustrated by superimposing at the leftmost end of the curve in the inset of the right panel of Fig. 6, the values of the critical value of M_T obtained for both finite and vanishing Soret coefficients, determined using the asymptotic analysis and fully numerical solution of the eigenvalue problem along with the reference value given by Eq. (35). Therefore, we stress that stratification of the thermal conductivity of the nanofluid and moderate values of the average concentration of nanoparticles Φ are the dominant factors for the onset of oscillatory thermocapillary instability. Finally, we remark that all the results obtained via the long-wave analysis have been checked against the numerical solution of the incompressible version of the EVP Eqs. (23) as in Ref. 28 and the agreement between them has been found to be excellent.

VI. SUMMARY AND CONCLUSIONS

This paper deals with the asymptotic analysis of the long-wave thermosolutal instabilities in a nanofluid layer with a deformable interface open to a gas phase in the gravity field. To begin with, we outline the mathematical model relevant to surface tension-driven instabilities in nanofluid layers, where a linear decrease in surface tension with the

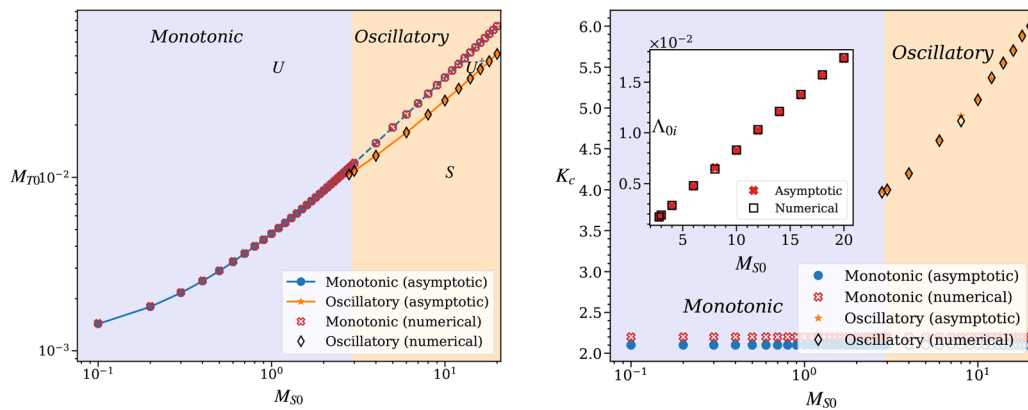


FIG. 5. Combined thermo-solutocapillary instability with $\Phi = 0.01$, $\eta = 0.31$, $G = 0.001$, $B = 10^{-8}$, $a = 7.47$, $L = 10^{-3}$, and $\Sigma_0 = 5 \times 10^3$. Left panel: variation of the long-wave critical curve represented in the plane $M_{S0} - M_{T0}$. Right panel: variation of the long-wave critical wavenumber K_c for the onset of oscillatory instability via the codimension-two point bifurcation; Inset: Variation of the critical frequency $\Lambda_{0i} = \Im(\Lambda_0)$ with the solutocapillary Marangoni number M_{S0} along the critical curve of the left panel. The $\circ$ and $\star$ points denote the threshold values of the long-wave monotonic and oscillatory instabilities, respectively, obtained using the asymptotic method of Sec. IV. The $\times$, $\diamond$, and $\square$ points represent the threshold values obtained via solution of the incompressible version of the EVP Eq. (23) as in Ref. 28. The symbols U , S , and U^t stand for the domains of monotonic instability, stability, and oscillatory instability of the system, respectively.

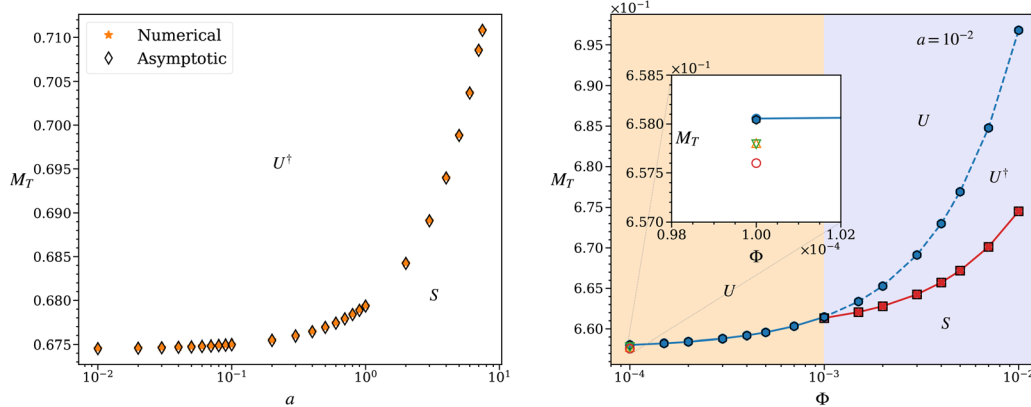


FIG. 6. Variation of the critical Marangoni number M_T of the pure long-wave thermocapillary instability at $\eta = 0.31$, $G = 1$, $B = 10^{-8}$, $L = 10^{-3}$, $M_S = 0$, and $\Sigma_0 = 1$. Left panel: Variation of M_T with the thermal conductivity stratification parameter a at $\Phi = 0.01$. The $\star$ and $\diamond$ points denote the critical values for the oscillatory thermocapillary instability obtained via numerical solution of the incompressible version of the EVP Eq. (23) as in Ref. 28 and the long-wave asymptotic expansion of Sec. IV, respectively. Right panel: Variation of M_T with the average bulk nanoparticle concentration Φ at $a = 0.01$. The $\times$ and $\circ$ points represent the results obtained via numerical solution of the EVP as in Ref. 28, whereas $\square$ and $\triangleright$ points represent the threshold values obtained via the long-wave expansion of Sec. IV, respectively. In the inset, the $\triangle$ and ∇ symbols represent, respectively, the results obtained from the fully numerical solution of the EVP and from the asymptotic theory for an almost vanishing Soret coefficient $\eta = 10^{-6}$, whereas the $\circ$ symbol depicts the reference value given by Eq. (35). The symbols U , $U^\dagger$, and S stand for the domains of oscillatory, monotonic, and stable domains of the system, respectively.

interfacial temperature and nanoparticle concentration is assumed. The set of governing equations expresses conservation laws of mass, momentum, energy, and nanoparticle concentration. We assume that nanoparticle mass flux is driven by two dominant mechanisms, namely, the Brownian diffusion and the Soret effect. We note that all the material properties of a nanofluid depend on the local particle concentration. A detailed account of modeling the thermophysical properties of a nanofluid can be found in Ref. 28. Similarly to what is done there, we assume linear dependencies of the density, specific heat, thermal conductivity, and the Soret coefficient, and fractional dependence of viscosity on particle concentration. We find the quiescent base state of the system, which is expressed via the Lambert W function. To investigate the stability of this base state, the eigenvalue problem in terms of the amplitudes of normal perturbations is derived. A full numerical investigation of the spectrum of this eigenvalue problem is quite costly, so any reasonable simplification of it is desirable.

The derived eigenvalue problem may be simplified based on the typically small values of the Lewis number $L \ll 1$. Under this assumption, the compressibility is small and the system may be treated as incompressible. Taking into account that our earlier results²⁸ showed the emergence of the long-wave thermo-solutocapillary instability using asymptotic analysis. Based on the general structure of the neutral Marangoni number,^{17,19} two distinct asymptotic limits are possible: (i) $B \sim O(\varepsilon^4)$, $k \sim O(\varepsilon)$, and $\Sigma_0 \sim O(1)$; (ii) $B \sim O(\varepsilon^2)$, $k \sim O(\varepsilon)$, and $\Sigma_0 \sim O(\varepsilon^{-2})$, where $\varepsilon \ll 1$ is a measure of the smallness of the relevant domain of wavenumbers k . We find that the second selection provides us with the needed information about the instability characteristics already at the second order of expansion.

We find that for $G \sim O(1)$, the oscillatory thermocapillary instability prevails over the monotonic one. Moreover, we reveal that the

critical values of the wavenumber and the frequency of the oscillatory thermocapillarity obey the scalings $k_c \sim \Sigma_0^{-1/4}$ and $\Im(\lambda) \sim \Sigma_0^{-1/2}$, respectively, with the critical Marangoni number increasing linearly with $\Sigma_0^{1/2}$. This suggests that the stronger the interfacial deformation, the smaller the distance between the depression and the substrate; hence, the system needs a lower Marangoni number to be destabilized. On the other hand, we find that for a fixed surface tension number, the critical wavenumber and the frequency increase with a small Biot number via $k_c \sim B^{1/4}$ and $\Im(\lambda) \sim B^{1/2}$, respectively. Remarkably, we find that the variation of the critical thermal Marangoni number for the oscillatory instability with both surface tension and Biot numbers is bilinear, in compliance with Eq. (28a). Interestingly, the oscillatory thermo-solutocapillary instability is borne via the codimension-two point with the critical wavenumber larger than the corresponding wavenumber of the monotonic instability for slightly lower values of both thermal and solutal Marangoni numbers. We also show that a combination of thermal conductivity stratification and a moderate nanoparticle concentration leads to the onset of the long-wave oscillatory instability. The critical thermal Marangoni number corresponding to the long-wave instability exhibits the transition from the oscillatory instability to the monotonic one by reducing the value of the average particle concentration Φ when the value of the thermal conductivity stratification parameter a is low.

To conclude, we emphasize that the long-wave asymptotic analysis presented here greatly facilitates an investigation of the thermo-solutocapillary instability in such a complex nanofluid system with excellent agreement with the results of a fully numerical investigation based on the EVP Eqs. (23) of Ref. 28. Furthermore, we show that the current analysis provides an additional and significantly less resource-demanding tool for the exploration of other long-wave regimes for most of the results obtained in Ref. 28 for a layer with a deformable interface heated at the substrate.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Raj Gandhi: Formal analysis (equal); Investigation (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Alexander Nepomnyashchy:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Alexander Oron:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: LONG-WAVE PROBLEM AT $O(1)$

Substituting Eqs. (29) and (30) into the eigenvalue problem (23) yields a sequence of boundary-value problems at orders ε^0 , ε^2 , etc. At leading order, the energy and nanoparticle concentration equations read

$$-\frac{\eta \mathcal{P}(a\phi_m \varphi_0 + (\mathcal{P} + 1)^2 \mathcal{D} \theta_0)}{(\mathcal{P} + 1)^3} + \frac{a\phi_m \mathcal{D} \varphi_0}{\mathcal{P} + 1} - (\mathcal{P} + 1) \mathcal{D}^2 \theta_0 = 0, \quad (\text{A1a})$$

$$\frac{\eta \left[(\mathcal{P} + 1)^2 \mathcal{D} \varphi_0 - \eta \mathcal{P} \varphi_0 - \frac{\mathcal{P}(\mathcal{P} + 1)^2 [(\mathcal{P} + 1) \mathcal{D}^2 \theta_0 + \eta \mathcal{D} \theta_0]}{a\phi_m} \right]}{(\mathcal{P} + 1)^3} - \mathcal{D}^2 \varphi_0 = 0, \quad (\text{A1b})$$

where $\mathcal{D} \equiv \frac{d}{dz}$, and $\mathcal{P} \equiv \mathcal{P}(z) = \mathcal{W}(\gamma(\Phi) a\phi_m \exp(\gamma(\Phi) a\phi_m + \eta z))$, which is associated with the Lambert function $\mathcal{W}$.

Equations (A1a) and (A1b) are rewritten in the matrix form with a zero eigenvector of the system $\mathbf{A}_0 \equiv \mathbf{A}_0(z) = \begin{pmatrix} \theta_0 \\ \varphi_0 \end{pmatrix}$ as

$$\mathcal{L} \mathbf{A}_0 = 0, \quad (\text{A1c})$$

where

$$\mathcal{L} = \begin{pmatrix} \left(\frac{-\mathcal{P}\eta}{\mathcal{P} + 1} \right) \mathcal{D} - (1 + \mathcal{P}) \mathcal{D}^2 & -\frac{\eta \mathcal{P} a\phi_m}{(\mathcal{P} + 1)^3} + \left(\frac{a\phi_m}{\mathcal{P} + 1} \right) \mathcal{D} \\ \left(-\frac{\eta^2 \mathcal{P}}{(\mathcal{P} + 1) a\phi_m} \right) \mathcal{D} - \left(\frac{\eta \mathcal{P}}{a\phi_m} \right) \mathcal{D}^2 & -\frac{\eta^2 \mathcal{P}}{(\mathcal{P} + 1)^3} + \left(\frac{\eta}{\mathcal{P} + 1} \right) \mathcal{D} - \mathcal{D}^2 \end{pmatrix}. \quad (\text{A1d})$$

The corresponding homogeneous boundary conditions at $O(1)$ are

$$(\mathcal{P}(0) + 1) \mathcal{D} \theta_0 - \frac{a\phi_m \varphi_0}{\mathcal{P}(0) + 1} = 0, \quad (\text{A1e})$$

$$\frac{\eta \mathcal{P}(0) \mathcal{D} \theta_0}{a\phi_m} - \frac{\eta \varphi_0}{\mathcal{P}(0) + 1} + \mathcal{D} \varphi_0 = 0 \quad \text{at } z = 0,$$

$$(\mathcal{P}(1) + 1) \mathcal{D} \theta_0 - \frac{a\phi_m \varphi_0}{\mathcal{P}(1) + 1} = 0,$$

$$\frac{\eta \mathcal{P}(1) \mathcal{D} \theta_0}{a\phi_m} - \frac{\eta \varphi_0}{\mathcal{P}(1) + 1} + \mathcal{D} \varphi_0 = 0 \quad \text{at } z = 1. \quad (\text{A1f})$$

We note that Eqs. (A1a)–(A1f) represent the leading order approximation for the original set of equations and boundary conditions of the eigenvalue problem (23). We find that the zero-order problem does not contain the Biot number B , which scales as $O(\varepsilon^2)$. This results in the degeneracy of the problem at hand with an

effective zero Biot number, and thereby leads to the emergence of the zero-order solution $\theta_0^{(1)} = 1$, $\varphi_0^{(1)} = 0$ associated with a thermal mode.

We find that there exists a second linearly independent solution for Eqs. (A1a)–(A1f) associated with a slowly evolving concentration mode, which can be obtained by differentiation of the steady base state of the system given by Eq. (17) with respect to the average volume concentration parameter $\gamma(\Phi)$. Therefore, a second solution for the zero-order problem is

$$\theta_0^{(2)} = \left(\frac{dT_0}{d\gamma(\Phi)} \right) = \frac{(\gamma(\Phi) a\phi_m + 1)(\mathcal{P} - \mathcal{P}(1))}{\eta \gamma(\Phi)(\mathcal{P} + 1)(\mathcal{P}(1) + 1)}, \quad (\text{A2})$$

$$\varphi_0^{(2)} = \left(\frac{d\phi_0}{d\gamma(\Phi)} \right) = \frac{(\gamma(\Phi) a\phi_m + 1) \mathcal{P}}{\gamma(\Phi) a\phi_m (\mathcal{P} + 1)}.$$

We carry on with the solution for the zero-order problem in the subspace spanned by its two nontrivial linearly independent solutions, namely,

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{u}_2 = \begin{pmatrix} \theta_0^{(2)} \\ \varphi_0^{(2)} \end{pmatrix}. \quad (\text{A3})$$

The general solution of the problem (A1a)–(A1f) is given by the superposition of two independent sets of solutions

$$\theta_0 = \alpha_1 \theta_0^{(1)} + \alpha_2 \theta_0^{(2)}, \quad \varphi_0 = \alpha_1 \varphi_0^{(1)} + \alpha_2 \varphi_0^{(2)}, \quad (\text{A4})$$

with two arbitrary constants α_1 and α_2 .

$$\mathcal{L}^\dagger = \begin{pmatrix} -\left(\frac{\eta \mathcal{P}}{\mathcal{P}+1}\right) \mathcal{D} - (1+\mathcal{P}) \mathcal{D}^2 & \left(-\frac{\eta^2 \mathcal{P}}{a\phi_m(\mathcal{P}+1)}\right) \mathcal{D} - \left(\frac{\eta \mathcal{P}}{a\phi_m}\right) \mathcal{D}^2 \\ -\left(\frac{a\phi_m}{\mathcal{P}+1}\right) \mathcal{D} & -\left(\frac{\eta}{\mathcal{P}+1}\right) \mathcal{D} - \mathcal{D}^2 \end{pmatrix}, \quad (\text{A5b})$$

and the boundary conditions obtained from the condition of a zero bilinear concomitant $\mathcal{J}(\mathbf{A}_0, \mathbf{A}_0^*)$

$$\frac{\eta \mathcal{P}(0) \mathcal{D} \varphi_0^*}{a\phi_m} + (\mathcal{P}(0)+1) \mathcal{D} \theta_0^* = 0, \quad \mathcal{D} \varphi_0^* = 0 \quad \text{at } z=0, \quad (\text{A5c})$$

$$\frac{\eta \mathcal{P}(1) \mathcal{D} \varphi_0^*}{a\phi_m} + (\mathcal{P}(1)+1) \mathcal{D} \theta_0^* = 0, \quad \mathcal{D} \varphi_0^* = 0 \quad \text{at } z=1. \quad (\text{A5d})$$

The simplified version of the adjoint problem

$$\mathcal{L}^\dagger \mathbf{A}_0^* = 0, \quad (\text{A6})$$

is given in Appendix A of Gandhi *et al.*²⁹ We note in passing that $\mathcal{L} \neq \mathcal{L}^\dagger$, which implies that the problem given by Eqs. (A1) is not self-adjoint, and the oscillatory instability may, in principle, occur.

We now determine the eigenfunctions of the adjoint problem given by Eqs. (A6), (A5c), and (A5d) corresponding to the zero eigenvalue of $\mathcal{L}^\dagger$. The analytical solution of the adjoint problem reads

$$\theta_0^* = c_1 z + c_2, \quad \varphi_0^* = c_3 \left(\ln(|\mathcal{P}|) - \frac{1}{\mathcal{P}} \right) - \frac{c_1 a \phi_m}{\eta} z + c_4, \quad (\text{A7})$$

where $c_1, \dots, c_4$ are constants, which are determined by substitution into the boundary conditions Eqs. (A5c) and (A5d) resulting in $c_1 = c_3 = 0$. Therefore, the zero eigenvectors of the adjoint problem (A5) are $\theta_0^* = c_2$ and $\varphi_0^* = c_4$, where c_2 and c_4 are arbitrary integration constants. We remark that the existence of two linearly independent solutions for the original problem at leading order suggests that the linear adjoint problem also possesses two linearly

We now proceed to derive the problem adjoint to Eqs. (A1a)–(A1f). Denote the adjoint eigenvector $\mathbf{A}_0^* = \begin{pmatrix} \theta_0^* \\ \varphi_0^* \end{pmatrix}$ and a scalar product of two eigenvectors $\mathbf{A}_0$ and $\mathbf{A}_0^*$ determined by $\langle \mathbf{A}_0^*, \mathbf{A}_0 \rangle = \int_0^1 dz (\theta_0^* \theta_0 + \varphi_0^* \varphi_0)$. In what follows, we use integration by parts to obtain the adjoint problem

$$\langle \mathbf{A}_0^*, \mathcal{L} \mathbf{A}_0 \rangle = \langle \mathcal{L}^\dagger \mathbf{A}_0^*, \mathbf{A}_0 \rangle + \mathcal{J}(\mathbf{A}_0, \mathbf{A}_0^*), \quad (\text{A5a})$$

where the linear adjoint operator is

independent solutions. Hence, in turn, we identify the second linearly independent solution associated with the concentration mode. Consequently, the linearly independent eigenvectors of the adjoint operator corresponding to its zero eigenvalue are

$$\mathbf{u}_1^* = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{u}_2^* = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (\text{A8})$$

The momentum conservation equation and the corresponding boundary conditions at zero order read

$$\begin{aligned} GK^2 \phi_m (\rho_{np} - 1) (\mathcal{P} + 1)^2 (\mathcal{P} - a\phi_m)^3 \varphi_0 \\ + \frac{2\eta^2 (a\phi_m)^2 \mathcal{P} [a\phi_m + \mathcal{P}(3\mathcal{P} + 2)]}{(\mathcal{P} + 1)(\mathcal{P} - a\phi_m)} \mathcal{D}^2 w_0 \\ + (\mathcal{P} + 1)(a\phi_m)^2 [(\mathcal{P} + 1)(\mathcal{P} - a\phi_m) \mathcal{D}^4 w_0 - 4\eta \mathcal{P} \mathcal{D}^3 w_0] = 0, \end{aligned} \quad (\text{A9a})$$

$$\text{at } z=0: w_0 = 0, \quad \mathcal{D} w_0 = 0, \quad (\text{A9b})$$

at the deformable interface, $z=1$:

$$\zeta_0 \Lambda_0 P - w_0 = 0, \quad (\text{A9c})$$

$$\begin{aligned} \frac{\eta \zeta_0 K^2 \mathcal{P}(1) M_{s0}}{(\mathcal{P}(1)+1) a \phi_m} + \frac{(a \phi_m)^2 \mathcal{D}^2 w_0}{(a \phi_m - \mathcal{P}(1))^2} \\ + K^2 \left[M_{T0} \left(\theta_0 - \frac{\zeta_0}{\mathcal{P}(1)+1} \right) + M_{s0} \varphi_0 \right] = 0, \end{aligned} \quad (\text{A9d})$$

with

$$\zeta_0 = \frac{(a \phi_m)^3 ((\mathcal{P}(1)+1) \mathcal{D}^3 w_0 (\mathcal{P}(1) - a \phi_m) - 2\eta \mathcal{P}(1) \mathcal{D}^2 w_0)}{K^2 (\mathcal{P}(1)+1) (\mathcal{P}(1) - a \phi_m)^3 (a \phi_m (G + K^2 \Sigma_s) + G \mathcal{P}(1) \phi_m (\rho_{np} - 1))}. \quad (\text{A9e})$$

REFERENCES

- ¹L. E. Scriven and C. V. Sternling, "The Marangoni effects," *Nature* **187**, 186 (1960).
- ²R. Levy, D. Uminsky, A. Park, and J. Calambokidis, "A theory for the hydrodynamic origin of whale flukeprints," *Int. J. Non-Linear Mech.* **46**, 616–626 (2011).
- ³D. Schwabe, A. Scharmann, F. Preisser, and R. Oeder, "Experiments on surface tension driven flow in floating zone melting," *J. Cryst. Growth* **43**, 305–312 (1978).
- ⁴M. F. Schatz and G. P. Neitzel, "Experiments on thermocapillary instabilities," *Annu. Rev. Fluid Mech.* **33**, 93–127 (2001).
- ⁵H. Greenspan, "On the dynamics of cell cleavage," *J. Theor. Biol.* **65**, 79–99 (1977).
- ⁶S. H. Davis, "Thermocapillary instabilities," *Annu. Rev. Fluid Mech.* **19**, 403–435 (1987).
- ⁷S. Ostrach, "Low-gravity fluid flows," *Annu. Rev. Fluid Mech.* **14**, 313–345 (1982).
- ⁸J. R. A. Pearson, "On convection cells induced by surface tension," *J. Fluid Mech.* **4**, 489–500 (1958).
- ⁹L. E. Scriven and C. V. Sternling, "On cellular convection driven by surface-tension gradients: Effects of mean surface tension and surface viscosity," *J. Fluid Mech.* **19**, 321–340 (1964).
- ¹⁰K. A. Smith, "On convective instability induced by surface-tension gradients," *J. Fluid Mech.* **24**, 401–414 (1966).
- ¹¹A. Oron and P. Rosenau, "Formation of patterns induced by thermocapillarity and gravity," *J. Phys. II* **2**, 131–146 (1992).
- ¹²R. J. Deissler and A. Oron, "Stable localized patterns in thin liquid films," *Phys. Rev. Lett.* **68**, 2948–2951 (1992).
- ¹³J. M. Burgess, A. Juel, W. D. McCormick, J. B. Swift, and H. L. Swinney, "Suppression of dripping from a ceiling," *Phys. Rev. Lett.* **86**, 1203–1206 (2001).
- ¹⁴V. Frumkin, W. Mao, A. Alexeev, and A. Oron, "Creating localized-droplet train by traveling thermal waves," *Phys. Fluids* **26**, 082108 (2014).
- ¹⁵M. C. Cross and P. C. Hohenberg, "Pattern formation outside of equilibrium," *Rev. Mod. Phys.* **65**, 851–1104 (1993).
- ¹⁶J. R. L. Skarda, D. Jacqmin, and F. E. McCaughan, "Exact and approximate solutions to the double-diffusive Marangoni–Bénard problem with cross-diffusive terms," *J. Fluid Mech.* **366**, 109–134 (1998).
- ¹⁷A. Oron and A. A. Nepomnyashchy, "Long-wavelength thermocapillary instability with the Soret effect," *Phys. Rev. E* **69**, 016313 (2004).
- ¹⁸S. W. Joo, "Marangoni instabilities in liquid mixtures with Soret effects," *J. Fluid Mech.* **293**, 127–145 (1995).
- ¹⁹A. Podolny, A. Oron, and A. A. Nepomnyashchy, "Long-wave Marangoni instability in a binary-liquid layer with deformable interface in the presence of Soret effect: Linear theory," *Phys. Fluids* **17**, 104104 (2005).
- ²⁰S. Shklyaev, A. A. Nepomnyashchy, and A. Oron, "Three-dimensional oscillatory long-wave Marangoni convection in a binary liquid layer with the Soret effect: Bifurcation analysis," *Phys. Fluids* **19**, 072105 (2007).
- ²¹S. Shklyaev, A. A. Nepomnyashchy, and A. Oron, "Marangoni convection in a binary liquid layer with the Soret effect at small Lewis number: Linear stability analysis," *Phys. Fluids* **21**, 054101 (2009).
- ²²S. Shklyaev and A. Nepomnyashchy, *Longwave Instabilities and Patterns in Fluids* (Birkhäuser, New York, NY, 2017).
- ²³M. Morozov, A. Oron, and A. A. Nepomnyashchy, "Long-wave Marangoni convection in a layer of surfactant solution," *Phys. Fluids* **26**, 112101 (2014).
- ²⁴B. Abramzon and W. Sirignano, "Droplet vaporization model for spray combustion calculations," *Int. J. Heat Mass Transfer* **32**, 1605–1618 (1989).
- ²⁵D. Lohse, "Fundamental fluid dynamics challenges in inkjet printing," *Annu. Rev. Fluid Mech.* **54**, 349–382 (2022).
- ²⁶J. Buongiorno, "Convective transport in nanofluids," *J. Heat Transfer* **128**, 240–250 (2006).
- ²⁷B. C. Pak and Y. I. Cho, "Hydrodynamic and heat transfer study of dispersed fluids with submicron metallic oxide particles," *Exp. Heat Transfer* **11**, 151–170 (1998).
- ²⁸R. Gandhi, A. Nepomnyashchy, and A. Oron, "Thermosolutal instabilities in a moderately dense nanoparticle suspension," *J. Fluid Mech.* **1011**, A52 (2025).
- ²⁹R. Gandhi, A. Nepomnyashchy, and A. Oron, "Longwave thermosolutal instability in a moderately dense nanoparticle suspension," *Eur. Phys. J. Spec. Top.* (to be published).
- ³⁰G. P. Krishnan, S. Beimfohr, and D. T. Leighton, "Shear-induced radial segregation in bidisperse suspensions," *J. Fluid Mech.* **321**, 371–393 (1996).
- ³¹M. Frank, D. Anderson, E. R. Weeks, and J. F. Morris, "Particle migration in pressure-driven flow of a Brownian suspension," *J. Fluid Mech.* **493**, 363–378 (2003).
- ³²A. Ramachandran and D. T. Leighton, "The influence of secondary flows induced by normal stress differences on the shear-induced migration of particles in concentrated suspensions," *J. Fluid Mech.* **603**, 207–243 (2008).
- ³³J. F. Morris, "Toward a fluid mechanics of suspensions," *Phys. Rev. Fluids* **5**, 110519 (2020).
- ³⁴H. Machrafi, "Surface tension of nanoparticle dispersions unravelled by size-dependent non-occupied sites free energy versus adsorption kinetics," *npj Microgravity* **8**, 47 (2022).
- ³⁵I. M. Krieger and T. J. Dougherty, "A mechanism for non-Newtonian flow in suspensions of rigid spheres," *Trans. Soc. Rheol.* **3**, 137–152 (1959).
- ³⁶S. H. Maron and P. E. Pierce, "Application of ree-eyring generalized flow theory to suspensions of spherical particles," *J. Coll. Sci.* **11**, 80–95 (1956).
- ³⁷C. G. de Kruif, E. M. F. van Iersel, A. Vrij, and W. B. Russel, "Hard sphere colloidal dispersions: Viscosity as a function of shear rate and volume fraction," *J. Chem. Phys.* **83**, 4717–4725 (1985). https://pubs.aip.org/aip/jcp/article-pdf/83/9/4717/18955035/4717_1_online.pdf.
- ³⁸J. C. Maxwell, *A Treatise on Electricity and Magnetism*, Cambridge Library Collection—Physical Sciences (Oxford University Press, 1873).
- ³⁹D. J. Jeffrey, "Conduction through a random suspension of spheres," *Proc. R. Soc. A* **335**, 355–367 (1973).
- ⁴⁰R. B. Bird, W. E. Stewart, and E. N. Lightfoot, *Transport Phenomena* (Wiley, New York, 2002).
- ⁴¹W. B. Russel, D. A. Saville, and W. R. Schowalter, *Colloidal Dispersions*, Cambridge Monographs on Mechanics (Cambridge University Press, 1989).
- ⁴²G. K. Batchelor, "Brownian diffusion of particles with hydrodynamic interaction," *J. Fluid Mech.* **74**, 1–29 (1976).
- ⁴³P. J. Whitmore and A. Meisen, "Estimation of thermo- and diffusiophoretic particle deposition," *Can. J. Chem. Eng.* **55**, 279–285 (1977).
- ⁴⁴K. I. Morozov, "On the theory of the Soret effect in colloids," in *Thermal Nonequilibrium Phenomena in Fluid Mixtures*, edited by W. Köhler and S. Wiegand (Springer Berlin Heidelberg, Berlin, Heidelberg, 2002), pp. 38–60.
- ⁴⁵M. Braibanti, D. Vigolo, and R. Piazza, "Does thermophoretic mobility depend on particle size?," *Phys. Rev. Lett.* **100**, 108303 (2008).
- ⁴⁶E. E. Michaelides, "Brownian movement and thermophoresis of nanoparticles in liquids," *Int. J. Heat Mass Transfer* **81**, 179–187 (2015).
- ⁴⁷C. V. Sternling and L. E. Scriven, "Interfacial turbulence: Hydrodynamic instability and the Marangoni effect," *AIChE J.* **5**, 514–523 (1959).
- ⁴⁸M.-H. Chang and A.-C. Ruu, "Rayleigh–Bénard instability in nanofluids: Effect of gravity settling," *J. Fluid Mech.* **950**, A37 (2022).
- ⁴⁹G. Buresti, "A note on Stokes' hypothesis," *Acta Mech.* **226**, 3555–3559 (2015).
- ⁵⁰R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, "On the LambertW function," *Adv. Comput. Math.* **5**, 329–359 (1996).
- ⁵¹G. Z. Gershuni and E. M. Zhukhovitskii, *Convective Stability of Incompressible Fluids* (Keter Publications, Jerusalem, 1976).
- ⁵²J. K. Platten and J. C. Legros, *Convection in Liquids* (Springer-Verlag, Berlin Heidelberg, 1984).
- ⁵³A. Podolny, A. A. Nepomnyashchy, and A. Oron, "Long-wave Marangoni instability in a binary liquid layer on a thick solid substrate," *Phys. Rev. E* **76**, 026309 (2007).
- ⁵⁴A. A. Nepomnyashchy, M. G. Velarde, and P. Colinet, *Interfacial Phenomena and Convection*, Monographs and Surveys in Pure and Applied Mathematics (CRC Press, 2001).
- ⁵⁵P. Colinet, J. C. Legros, and M. G. Velarde, *Nonlinear Dynamics of Surface-Tension-Driven Instabilities* (Wiley, Berlin, 2001).
- ⁵⁶S. Shklyaev and A. A. Nepomnyashchy, "Longwave Marangoni convection in a surfactant solution between poorly conducting boundaries," *J. Fluid Mech.* **718**, 428–456 (2013).
- ⁵⁷A. Nayfeh, *Introduction to Perturbation Techniques*, Wiley Classics Library (Wiley, New York, 2011).

- ⁵⁸S. Pushpavanam, *Mathematical Methods in Chemical Engineering, Eastern Economy Edition* (PHI Learning, New Delhi, 1998).
- ⁵⁹A. Varma and M. Morbidelli, *Mathematical Methods in Chemical Engineering* (Oxford University Press, Oxford, 1997).
- ⁶⁰J. N. Kutz, “Advanced differential equations: Asymptotics & perturbations,” [arXiv:2012.14591](https://arxiv.org/abs/2012.14591) [math.CA] (2020).
- ⁶¹S. H. Davis, “Rupture of thin liquid films,” in *Waves on Fluid Interfaces*, edited by R. E. Meyer (Academic Press, 1983), pp. 291–302.
- ⁶²A. Oron, S. H. Davis, and S. G. Bankoff, “Long-scale evolution of thin liquid films,” *Rev. Mod. Phys.* **69**, 931–980 (1997).
- ⁶³H. R. Brand, P. C. Hohenberg, and V. Steinberg, “Codimension-2 bifurcations for convection in binary fluid mixtures,” *Phys. Rev. A* **30**, 2548–2561 (1984).